Principal Component Analysis (PCA) – An overview

Introduction

Principal Component Analysis (PCA) is a technique used for dimensionality reduction and data visualization. It transforms high-dimensional data into a lower-dimensional data while preserving the patterns. PCA is commonly used to analyse data sets where visualizing relationships in higher dimensions is challenging.

The Need for PCA

When dealing with datasets containing numerous variables (e.g., gene measurements, weather readings), visualizing relationships becomes complex. For instance:

- With two variables, data can be plotted on a 2D graph but with four or more variables, visualizing relationships requires higher dimensions, which is not feasible.

PCA addresses this challenge by reducing the dimensionality of the data while retaining the variation. For example, PCA can transform a dataset with four variables into a 2D PCA plot that not only simplifies visualization but also highlights the most important components for clustering and analysis by introducing the best fitting line , known as the principal component.

Key Features of PCA

1. **Dimensionality Reduction**:
    - PCA reduces the number of dimensions by identifying new axes (principal components) that best represent the variation in the data.
2. **Preservation of Important Information**:
    - It prioritizes variables with greater significance, enabling effective clustering and interpretation.
3. **Evaluation of Accuracy**:
    - PCA provides metrics to assess how well the reduced-dimensionality representation captures the original data.

Understanding the Process

1. **Scaling of data**:

    It helps in visually interpret or analyse data effectively. The data is made roughly equivalent otherwise it will be biased towards one data**.** The most common method used for scaling is diving by the standard deviation of the provided data values.

2. **Centering the Data**:

- PCA starts by finding the centre point (mean) of the dataset and shifting the data centre such that the centre becomes the origin.
- Cantering ensures that variations in the data are measured relative to a common point.
- After centering , the average value for each gene will be 0 and after scaling , the standard deviation for the values for each gene will be 1.

3. **Clustering of Correlated Data**:
   - Highly correlated variables are grouped together, helping in identifying patterns and clusters.
4. **Finding the Best-Fitting Line (Principal Component)**:
   - PCA identifies the line that minimizes the distances between the data points and the line or maximizes the sum of squared distances from the origin to the projected points.
   - This line, called the **Principal Component (PC1)**, represents the direction of maximum variance in the data.
5. **Using Pythagoras' Theorem**:
   - The squared distance from a data point to the origin (a^2) equals the sum of the squared distances along the x-axis (b^2) and y-axis (c^2):
     - $\mathbf{a^2 = b^2 + c^2}$
   - PCA maximizes b^2 (distance along the principal component), thereby reducing c^2 (distance to the component).
   - Since a^2 (distance from point to origin) remains constant, increasing b^2 requires reducing c^2. PCA optimizes this balance to find the best-fitting line.
6. **Maximizing the Sum of Squared Distances**:
   - PCA finds the best-fitting line by maximizing the sum of squared distances (SS) from the projected points to the origin.
   - The line with the largest SS is called the principal component or PC1.
   - For instance, if the slope of PC1 is 0.25, it means moving 4 units along the x-axis corresponds to moving 1 unit along the y-axis. This indicates that the x-axis variable is more important than the y-axis variable. This relationship is referred to as a linear combination of data sets.
7. **Eigenvalues and Eigenvectors**:
   - The principal component is associated with an eigenvector, which is a unit vector (length = 1) representing its direction (making a=1)

- The eigenvalue corresponds to the sum of squared distances (SS) from the origin to the points projected on the line.
- Larger eigenvalues indicate more significant components.

8. **Proportion of Variance (Loading Scores)**:
   - Loading scores indicate the contribution of each variable to the principal component.
   - The square root of the SS is referred to as the singular value.

Visualizing PCA Results

1. **Selecting Principal Components**:
   - PCA calculates multiple principal components, each orthogonal (perpendicular) to the others.
   - The two components whose sum is accounting for the highest variance are typically selected for visualization.
2. **Adjusting Axes**:
   - The selected components are displayed as parallel to x and y axes to enhance readability.
3. **Plotting the Data**:
   - Data points are projected onto the principal components to create a 2D PCA plot.

Importance of Scaling in PCA

1. **Ensuring Fair Representation**:
   - Scaling ensures that all variables contribute equally to the PCA analysis. Without scaling, variables with larger magnitudes could dominate the results.
2. **Method for Scaling**:
   - Data is typically scaled by dividing each value by its standard deviation.
3. **Centered and Scaled Data**:
   - After centering, the mean value for each variable becomes zero.
   - After scaling, the standard deviation for each variable is set to one.

Practical Implementation Using Code

To perform PCA efficiently, preprocessing is essential:

1. **Using Scikit-learn for Scaling**:

    - The preprocessing package from Scikit-learn provides functions for scaling and centering data.

Example:

```
from sklearn.preprocessing import StandardScaler

scaled_data = StandardScaler().fit_transform(data.T)
```

    - Transposing (data.T) is necessary because Scikit-learn expects samples to be rows, not columns.

2. **Variation Calculation**:
    - Variation in python is calculated as:

```
(measurement - mean) ** 2 / number_of_measurements
```

3. **Reusable Objects**:
    - Scikit-learn uses objects (e.g., StandardScaler) that can be trained on one dataset and applied to others, enhancing flexibility.

Summary

Principal Component Analysis (PCA) is a handy tool for simplifying complex datasets. It helps reduce the number of variables while still capturing the most important information. By scaling and centering the data, PCA makes sure that all variables are treated equally, allowing us to identify meaningful patterns and groupings. With tools like Scikit-learn, you can apply PCA to your dataset quickly and efficiently